PHASE 02 / PROJECT DOCUMENTATION

Requirement Analysis

Converting stakeholder needs into data, feature, and quality requirements.

This phase specifies what the EV dashboard should do, which data it needs, and how the team will decide whether the delivered experience is useful and reliable.

01 | Stakeholder requirements

Stakeholder	Question / need	Expected support
Market Strategy Director	Which segments and manufacturers appear strongest?	Overview KPIs, segment comparison, manufacturer ranking
Business Intelligence Manager	How is performance distributed across companies and places?	Time trends, share calculations where valid, country and battery views
Sustainable Mobility Analyst	What patterns appear across battery types, range, and ratings?	Specification comparisons and transparent data coverage

02 | Functional requirements

- FR-01: Display the project purpose, dataset scope, and key metric definitions.
- FR-02: Summarize available records with KPIs that are calculated from real fields.
- FR-03: Compare manufacturers and vehicle categories using appropriate charts.
- FR-04: Explore battery types, capacity, and range when the fields exist.
- FR-05: Show sales over time and market share only when the dataset supports valid time-based totals and denominators.
- FR-06: Provide relevant filters and ensure connected views update consistently.
- FR-07: Include source details and explain known data-quality or coverage limitations.

03 | Quality requirements

Area	Requirement
Usability	Use readable titles, units, legends, tooltips, and consistent terminology.
Accuracy	Validate calculations independently; do not silently replace missing values with zero.
Performance	Record actual load and interaction behavior on the target device.
Maintainability	Keep a documented preparation workflow and clear metric definitions.
Transparency	Separate observed results from interpretation and avoid unsupported causal claims.
Responsible sharing	Check dataset licensing before committing source files to a public repository.

04 | Data requirements

Field, if present	Why it matters	Validation needed
Sales / units sold / year	Trend and contribution analysis	Period, unit, duplicates, comparable coverage
Manufacturer / model / category	Company and segment comparison	Consistent labels and category definitions
Battery type / capacity	Technology distribution and specification comparison	Normalize names; confirm capacity unit
Driving range	Compare expected range across segments	Standardize km/miles and inspect outliers
Country / region	Geographic distribution	Clarify whether field means manufacturing or sales location
Customer rating	Customer response comparison	Confirm rating scale and missing-value treatment

05 | Proposed data flow

Source file → profiling → cleaning and validation → metric calculations → visual analysis → interactive dashboard/story → findings and recommendations.

06 | Acceptance checklist

- Each visual has a defined question, data field, aggregation, and unit.
- Filter combinations produce consistent results across intended charts.
- Source and data coverage are documented in the dashboard or repository.
- Missing or unavailable fields are disclosed rather than simulated.
- A reviewer can understand how to open or reproduce the project.